AI & Language-Model APIs

Week 13 · APIs module · Textbook Chapter 13

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This week at a glance

An LLM is just another API — authenticate, send a request, parse the response. We use it to turn the messy text of Chapters 6–9 into structured data.

Monday | Concepts & notebook intro

- Chat completions, prompting for structure, embeddings, and the reproducibility catch

Wednesday | Notebook lab

- Classify, extract, and cluster Colorado bills across two providers

Friday | Textbook revisions

- Propose & peer-review pull requests improving Chapter 13 — with AI disclosure

Companion notebook

`ch-13-ai-apis.ipynb`

Bring a laptop

And your API keys (or the class sandbox key). Calls cost real money — we estimate first.

1. Monday · Concepts

LLMs are another API to learn

Large language models are a new category of tool for web data science. In Chapters 6–9 you collected **unstructured text** from web pages, PDFs, and archived documents. These models turn that text into structured data that you can analyze.

Typical jobs: sentiment analysis, entity extraction, content coding, and summarization — **at scale**.

The same three moves you already know apply: **authenticate**, **construct a request**, **parse the response**.

But LLMs are different

Three properties set them apart from a REST API:

- **Non-determinism** — the same input can give different output
- **Cost** — you pay per token
- **Reproducibility** — outputs drift as models change

Textbook Ch. 13, "LLMs as Web Data Science Tools." Builds on the scraping of Ch. 6–9.

Anatomy of a chat request

A chat request is a **list of messages**, each with a role:

- `system` — sets the assistant's behavior
- `user` — your input
- `assistant` — lets you seed prior turns

You get back a list of choices; the text lives at `response.choices[0].message.content`.

Two providers, near-identical shape: OpenAI's `chat.completions.create` and Anthropic's `messages.create` (where `system` is its own argument).

Messages in, one completion out.

Textbook Ch. 13, "The OpenAI API" & "The Anthropic API."

From free text to structured data

For research you rarely want prose — you want **fields you can put in a DataFrame**. Two techniques get you there:

Few-shot prompting

Show the model examples of the input→output format you want, then ask it to continue the pattern:

```
Summary: "…teacher salaries."
Category: education
Summary: "{new bill}"
Category: ___
```

Set `temperature=0` to make classification as repeatable as possible.

Let the API enforce the schema

Structured Outputs accept a JSON Schema and *guarantee* the response conforms — no more brittle parsing. Lighter-weight JSON mode guarantees valid JSON without a fixed schema; Anthropic gets there via **tool use**.

Beyond extraction, **embeddings** turn text into vectors so you can measure *semantic* similarity — more on Wednesday.

Textbook Ch. 13, "Structured Output and Few-Shot Prompting" & "Structured Outputs."

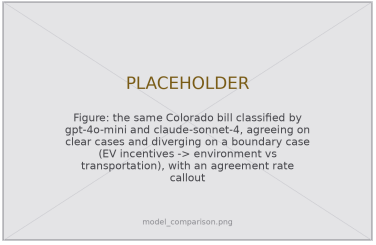
The catch: non-determinism, cost & reproducibility

When you parse HTML, the same input always gives the same output. LLMs **break that guarantee**:

- Even at `temperature=0`, outputs vary slightly (floating-point non-determinism)
- Model **weights change** behind a fixed name over time
- Retired models make exact replication **impossible**

Send the same prompt to two models and they mostly agree — but **disagree at the boundaries** (is an EV-incentive bill `environment` or `transportation`?). That is genuine ambiguity, not a bug.

Treat the model as a measurement instrument — and calibrate it against human judgment.



Two models, one prompt — agreement is a finding, not a given.

Textbook Ch. 13, "Reproducibility and Documentation" & "Comparing Model Outputs." Connects to the book's AI-disclosure appendix.

2. Wednesday · Notebook Lab

Setup & your first call

Store the key in the environment (never in code), then make one request:

```python
import os
from openai import OpenAI

# The key lives in an environment variable -- never in the source
client = OpenAI(api_key=os.environ.get("OPENAI_API_KEY"))

response = client.chat.completions.create(
    model="gpt-4o-mini",
    messages=[
        {"role": "system", "content": "You are a helpful assistant."},
        {"role": "user", "content": "What is web scraping in one sentence?"},
    ],
)
answer = response.choices[0].message.content # the model's reply text
```

Textbook Ch. 13, "Basic Chat Completion." Key storage: *Missing Manual* Ch. 29.

Few-shot classification, one bill at a time

Wrap the call in a function; keep the `sleep` in the *caller*:

```
def classify_bill(summary, client):
    response = client.chat.completions.create(
        model="gpt-4o-mini",
        messages=[
            {"role": "system", "content":
                "Classify this bill into ONE category: education, healthcare, "
                "environment, taxation, criminal_justice, ... or other. "
                "Respond with only the category label."},
            {"role": "user", "content": summary},
        ],
        temperature=0, # reduce randomness for classification
    )
    return response.choices[0].message.content.strip().lower()

for bill in bills: # each bill is a dict with a "summary"
    bill["category"] = classify_bill(bill["summary"], client)
    time.sleep(1) # rate-limit between API calls
```

Textbook Ch. 13, "Applied Example: Legislative Bill Analysis," Step 1.

Get JSON out — and stop it from breaking

The defensive fallback treats bad JSON as a fact of life:

```
# Fallback: ask for JSON, then parse defensively
try:
    extracted = json.loads(response.choices[0].message.content)
except json.JSONDecodeError:
    print("Model did not return valid JSON") # log it and move on
```

Better — hand the API a schema and it **guarantees** the shape:

```
response = client.chat.completions.create(
    model="gpt-4o-mini",
    messages=[{"role": "user", "content": bills[0]["summary"]}],
    response_format={"type": "json_schema", "json_schema": bill_schema},
    temperature=0,
)
result = json.loads(response.choices[0].message.content) # always parses
```

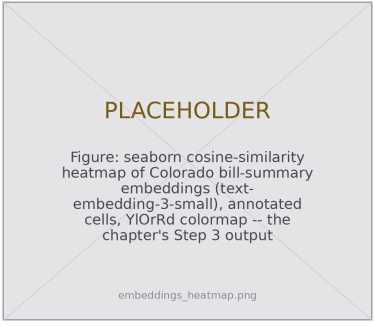
Anthropic reaches the same guarantee through **tool use**. Keep the `try/except` for models that offer no guarantee.

Textbook Ch. 13, Step 2 & "Structured Outputs: Letting the API Enforce the Format."

Embeddings: meaning as vectors

```python
def get_embeddings(texts, client,
                   model="text-embedding-3-small"):
    vecs = []
    for text in texts:
        r = client.embeddings.create(
            input=text, model=model)
        vecs.append(r.data[0].embedding)
        time.sleep(0.5) # rate limit
    return np.array(vecs)

emb = get_embeddings(summaries, client)
# 1.0 == same meaning, 0.0 == unrelated
sim = 1 - cosine(emb[i], emb[j])
```



Bills about “school funding” and “educational resources” score high despite no shared keywords.

Textbook Ch. 13, “Text Embeddings” & Step 3. Internals: *Missing Manual* Ch. 36.

Two providers, one prompt

Send the *identical* classification prompt to Claude and compare:

```
from anthropic import Anthropic
client_a = Anthropic(api_key=os.environ.get("ANTHROPIC_API_KEY"))

# Anthropic: `system` is its own argument; text is response.content[0].text
resp = client_a.messages.create(
    model="claude-sonnet-4-20250514",
    max_tokens=50,
    system=system_prompt,
    messages=[{"role": "user", "content": bill_text}],
)
anthropic_label = resp.content[0].text.strip()

print(openai_label, anthropic_label, openai_label == anthropic_label)
```

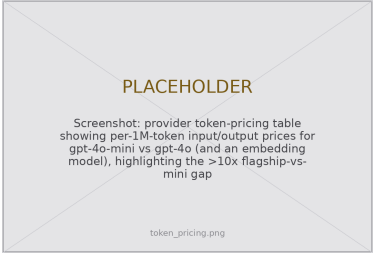
Agreement on 85% of cases is common; the 15% of disagreements are exactly where you need a human-labeled sample to calibrate.

Textbook Ch. 13, "The Anthropic API" & "Comparing Model Outputs."

Mind the cost, pin the version

```python
# Estimate BEFORE a big batch.
# gpt-4o-mini ~ $0.15 in / $0.60 out per 1M tok
n_bills = 500
in_tokens = n_bills * 200 # summary ~200 tok
out_tokens = n_bills * 5 # label ~5 tok
cost = (in_tokens*0.15 + out_tokens*0.60) / 1e6
print(f"Estimated cost: ${cost:.4f}")
```

For research, pin the **dated** model (`gpt-4o-mini-2024-07-18`) and log every call — model, full prompt, params, timestamp — to a JSONL audit trail.



Flagship models cost $>10\times$ their mini siblings. Develop on mini; promote to the big model only after validating prompts.

Textbook Ch. 13, “Cost Management” & “Reproducibility and Documentation.”

Lab: work these, then show-and-tell

Work in pairs. Do these exercises from the notebook:

1. Extract structured data from 10 or more documents that you collected. Design the method, test it, and assess the quality.
2. Cluster 20 or more documents by cosine similarity. Report if the clusters agree with your expectation.
3. Code 20 items manually. Then let a language model classify them. Report the items where the two results disagree.
4. Compare two models on 20 items that you coded manually.
5. Make a dendrogram of 30 Wikipedia titles from three topics.
6. **5617**: do a validation study. Use 50 items and code 25 of them manually. Report Cohen's κ. Compare your result with Ziems et al. (2024)

Show-and-Tell

Bring one of these to class:

- a prompt that surprised you (good or bad)
- data that you extracted and find interesting
- a research provocation about LLMs as instruments

An LLM is a measurement instrument, not an oracle.
Same prompt, different model — different answer.

Pin the version, log the call, validate against a human.

Report *what you ran*, not just what you found.

3. Friday · Textbook Revisions

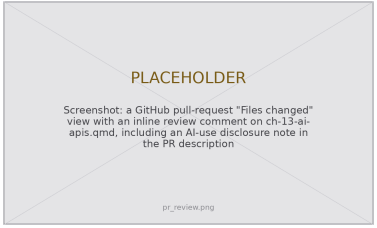
Improving Chapter 13 together

Today we make the book better. Each of you opens a **pull request** against `ch-13-ai-apis.qmd` and reviews a classmate's.

The workflow (from Week 1):

1. Branch / fork the book repo
2. Edit the chapter; commit with a clear message
3. Open a PR describing *what* and *why*
4. Review two peers' PRs: kind, specific, actionable

This chapter is *about* AI. If a language model helped you to revise it, **disclose** that help. The book asks for this disclosure.



Contributions & reviews count toward the Textbook Revisions grade (15%).

A revision menu for Chapter 13

High-value targets — pick one and scope it to a single PR:

- **Re-verify the model names.** The chapter's own "Model Names Rotate" callout warns these go stale. Check the providers' current lineups; update names or add a dated note.
- **Deepen "Common Issues to Debug."** Add a worked API-key-in-git-history recovery, or a real JSON-parse failure and its `try/except` fix.
- **Add a figure the prose implies.** A request/response diagram, the token-pricing comparison, or the embedding-similarity heatmap would all help.
- **Contribute a reproducibility helper.** The chapter recommends logging each call to JSONL — ship a small reusable `log_call()` function.
- **Strengthen a thin exercise.** Add expected output, a rubric, or a starter cell so an exercise is self-checking.
- **Extend "Further Reading."** An open-weight-models entry (LLaMA, Mistral, Qwen) tied to the access-inequality callout.

Targets drawn from the chapter's callouts, "Common Issues to Debug," and "Further Reading."

A good revision, a good review — and honest disclosure

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Renders (`quarto render`) cleanly
- Matches the book's voice

A strong review

- Runs the code, not just skims
- Names specifics; splits must-fix from nice-to-have
- Ends with a clear verdict

Disclose your AI use

The book's **AI Coauthorship & Responsible Disclosure** appendix asks four questions — answer them in your PR when AI helped:

1. **What tool?** model & version (e.g. GPT-4o, March 2026)
2. **What task?** drafting prose, generating a code cell, classifying text…
3. **What oversight?** how you tested & verified the output
4. **What limitations?** where you are unsure

Disclosure is about *transparency, not permission* — and you remain fully accountable for what you submit.

Next week: **Automating data collection —** schedulers, pipelines, and keeping a scraper alive.

Key takeaways

1. An LLM is **another API**: authenticate, construct a request, parse the response — read the docs and manage your keys.
2. Prompt for **structure**: few-shot examples, `temperature=0`, and schema-enforced outputs beat parsing free text.
3. **Embeddings** measure semantic similarity where keywords fail — classify, cluster, and compare at scale.
4. Non-determinism, cost, and reproducibility are **real**: estimate spend, pin dated model versions, and log every call.
5. Treat the model as a **measurement instrument** — validate against human judgment and **disclose** how AI was used.

Reading: Textbook Ch. 13. Related: Ziems et al. (2024); OpenAI & Anthropic API docs. Disclosure: the AI Coauthorship appendix.